

Selection Sort

```
void selection-sort (int arr[], int size)
{
```

```
    for (int i=0; i<size-1; i++)
    {
```

```
        int smallest=i;
```

```
        for (int j=i+1; j<size; j++)
        {
```

```
            if (arr[j]<arr[smallest])
```

```
                smallest=j;
```

```
        }
```

```
        int temp = arr[i];
```

```
        arr[i] = arr[smallest];
```

```
        arr[smallest] = arr temp;
```

```
    }
```

```
}
```

Properties:

ct con

[h Stable

/n&/nd or

iv) Structure Dependant.

v) worst & Best case Complexity $O(n^2)$

it) inplace W)



s mqLlost=• L		
synqLQost=		<u>smallest=4</u>
smûest=	st	<u>smallest=5</u>
sm ouest=		QLLest = 5
S grnaLQesœ\		<u>smallest=5</u>
Ste <u>smallest=1</u>	ste	-g srTuLleC= S ₂
s e tzl		<u>1,2,2,3,8 9,4,6</u>
<u>smallest=1</u>	asss:	snzEegT=Q •zS
S toa s rnQQ-LejT		est
alla <u>1 2 9 9 8 3 4 6 2</u>		eff= 4
Pass <u>i=1, smallest=1</u>		<u>smallest=6</u>
sce g moues t =	St	<u>smallest=6</u>
smcd)-Qest		<u>1,2,2,3,4,9,8,6,9</u>
est—l		Es mal/est=S
S s°rn atz)	ste	Ilest-
spnaest=		—7 no lest=7
synoJUe/tzJ	st	<u>smallest=7</u>
<u>i=8, smallest=1</u>		<u>1,2,2,3,4,6,8,9,9</u>
LDa <u>smallest 8</u>		Pgs70 —6 smalest=6
e ' 2 q q g 3 q 6 2	ste	<u>smallest=6</u>
Pass3: snoadZes&2		<u>smallest=6</u>
<u>step1:</u> <u>smallest=2</u>		<u>1,2,2,3,4,6,8,9,9</u>
20 Sm //est2 L/		m llesT=D
<u>step3:</u> smal/ sta	st	<u>smallest=7</u>

sm	0 1 2 2 3 q. <u>6.8.9.9</u>
st <u>smallest=5</u>	<u>7.7</u>
step 6: $j=8$, $smallest=8$ swap $smallest$ & i array: 1, 2, 2, 9, 8, 3, 4, 6, 9 2 7 7 7	
Pass 1: $i=0$, $smallest=0$, $j=1,8$ step 1: $i=1$ $smallest=0$ Pass 4: $i=3$, $smallest=3$, $j=4,8$	



input array: 9, 2, 9, 1, 8, 3, 4, 6, 2

Pass 1: 1, 2, 9, 9, 8, 3, 4, 6, 2

Pass 2: 1, 2, 9, 9, 8, 3, 4, 6, 2

Pass 3: 1, 2, 2, 9, 8, 3, 4, 6, 9

Pass 4: 1, 2, 2, 3, 8, 9, 4, 6, 9

Pass 5: 1, 2, 2, 3, 4, 9, 8, 6, 9

Pass 6: 1, 2, 2, 3, 4, 6, 8, 9, 9

Pass 7: 1, 2, 2, 3, 4, 6, 8, 9, 9

Pass 8: 1, 2, 2, 3, 4, 6, 8, 9, 9